



Industrial Internship Report on

“Autonomous Crop and Weed Detection in Sesame Agriculture Using YOLOv8”

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Executive Summary

This report documents the Industrial Internship undertaken through UpSkill Campus (USC) and The IoT Academy, in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). The internship spanned six weeks and centred on Project 5: Autonomous Crop and Weed Detection for Sesame Agriculture using Deep Learning. The project required building an end-to-end computer vision pipeline — from raw field image collection and annotation through to real-time inference — within the stipulated timeline.

The core deliverable is a trained YOLOv8 object detection model capable of distinguishing sesame crop plants (class 0) from weed species (class 1) in real time, producing bounding-box predictions suitable for integration with a robotic precision-spraying actuator. A dataset of 1,300 annotated 512 × 512 RGB images was curated through field photography, quality filtering, resolution normalisation, and class-preserving data augmentation. The model was fine-tuned via transfer learning on a Google Colab T4 GPU, achieving inference latency of approximately 2–5 ms per frame. This internship provided substantial exposure to industrial deep learning workflows, dataset engineering, and deployment-oriented model design.

GitHub Link: <https://github.com/atharva481/crop-weed-detection>

Report Link: https://github.com/atharva481/crop-weed-detection/CropWeedDetection_Atharva_USC_UCT.pdf

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1 Preface

This report summarises six weeks of industrial internship activity undertaken as part of the UpSkill Campus (USC) and The IoT Academy programme, executed in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship was structured around a real-world problem statement assigned by UCT: the design and implementation of an autonomous crop-versus-weed detection system for sesame agriculture using state-of-the-art deep learning techniques.

The relevance of industry-aligned internships to career development in the engineering and data science domains cannot be overstated. Academic curricula, while theoretically rigorous, seldom expose students to the constraints that characterise production-grade systems: dataset imperfections, compute budgets, latency requirements, and iterative debugging under time pressure. This internship bridged that gap by requiring the design of a complete machine learning pipeline — from raw data collection to trained-model inference — within a fixed six-week window.

The project centred on building a binary object detection model capable of real-time discrimination between sesame crop plants and weed species using the YOLOv8 single-stage detection framework. The dataset of 1,300 annotated field images was curated, preprocessed, and augmented from 589 raw photographs; the model was trained on a Google Colab T4 GPU and validated against a held-out set of 260 images.

The opportunity afforded by USC and UCT to work on a genuine industrial problem — one with direct implications for herbicide waste reduction and crop contamination prevention — was both challenging and immensely rewarding. The programme was structured into weekly milestones: dataset preparation (Weeks 1–2), model architecture study and baseline training (Week 3), hyperparameter tuning and evaluation (Week 4), inference pipeline development (Week 5), and report writing and code submission (Week 6).

I am sincerely grateful to the mentors and coordinators at UCT and USC whose guidance shaped the direction of this work, and to my peers who provided constructive feedback throughout. To fellow students undertaking this programme: embrace the ambiguity of industrial problems — the ability to convert an underspecified problem statement into a working technical solution is the most transferable skill you will develop.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

UniConverge Technologies Pvt Ltd (UCT) is a digital transformation company established in 2013, specialising in industrial solutions with a prime focus on sustainability and return on investment (RoI). UCT develops and deploys products and solutions leveraging cutting-edge technologies including Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, and Python-based backends.

Key product lines offered by UCT include:

- UCT Insight (IoT Platform) — A Java/ReactJS-based IoT platform supporting device connectivity via MQTT, CoAP, HTTP, Modbus TCP, and OPC UA protocols. It provides dashboard building, analytics, alert management, third-party integration (Power BI, SAP, ERP), and a rule engine for both cloud and on-premises deployments.
- FactoryWatch (Smart Factory Platform) — A scalable production and asset monitoring solution offering OEE tracking, predictive maintenance, and digital twin capabilities delivered via a modular SaaS architecture.
- LoRaWAN-based Solutions — UCT is an early adopter of LoRaWAN technology, providing solutions for Agritech, Smart Cities, Industrial Monitoring, Smart Street Lighting, and utility metering.
- Predictive Maintenance — Industrial machine health monitoring and remaining useful life (RUL) estimation using embedded systems, Industrial IoT, and machine learning.

2.2 About UpSkill Campus (USC)

UpSkill Campus (USC), in association with The IoT Academy and UniConverge Technologies, facilitates the end-to-end execution of the industrial internship programme. USC is a career development platform delivering personalised executive coaching in a scalable, measurable, and affordable format. By integrating self-paced learning with support services including internships, live projects, and industry-expert interactions, USC aims to upskill one million learners within five years.

The IoT Academy is the EdTech division of UCT, conducting executive certification programmes in collaboration with EICT Academy, IIT Kanpur, IIT Roorkee, and IIT Guwahati across multiple emerging technology domains.

2.3 Objectives of this Internship Programme

The objectives defined for this internship were:

- ☑ To gain practical experience of working within an industry-grade project environment with real-world constraints.
- ☑ To solve an authentic industrial problem — herbicide waste and crop contamination in sesame farming — using deep learning.
- ☑ To improve employment prospects through demonstrable project experience and a documented technical deliverable.
- ☑ To develop a deeper applied understanding of computer vision, dataset engineering, and model deployment.
- ☑ To cultivate professional competencies including technical communication, time management, and independent problem-solving.

2.4 References

- [1] Jocher, G., Chaurasia, A., & Qiu, J. (2023). Ultralytics YOLOv8. <https://github.com/ultralytics/ultralytics>
- [2] Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection. IEEE CVPR.
- [3] Oerke, E. C. (2006). Crop losses to pests. The Journal of Agricultural Science, 144(1), 31–43.
- [4] Lin, T. Y. et al. (2014). Microsoft COCO: Common Objects in Context. ECCV.
- [5] UpSkill Campus: <https://www.upskillcampus.com/>

2.5 Glossary

Term / Full Form	Acronym
You Only Look Once	YOLO
Convolutional Neural Network	CNN
Mean Average Precision	mAP
Intersection over Union	IoU
Graphics Processing Unit	GPU
Non-Maximum Suppression	NMS
Red Green Blue (colour image)	RGB
Data Augmentation	DA
Transfer Learning	TL
Microsoft Common Objects in Context	MS COCO

3 Problem Statement

In the assigned problem statement, the intern was tasked with developing a real-time computer vision system for the autonomous detection and classification of crop plants and weeds in sesame (*Sesamum indicum*) agricultural fields.

Weed infestation is one of the most economically significant biotic constraints in global crop production. In sesame cultivation, competing weed species consume soil nutrients, water, and light that would otherwise support crop growth, leading to yield reductions of 10–40% in severe infestation scenarios. The conventional farmer response — uniform field-wide broadcast spraying of herbicides — is both chemically wasteful and potentially harmful: pesticide residues can adhere to sesame capsules (the harvested organ), degrade soil microbial diversity, and contribute to surface-water contamination through agricultural runoff.

The specific challenge is formulated as a two-class, real-time object detection problem: given an RGB image captured from a drone or ground-robot camera platform flying over a sesame field, the system must produce axis-aligned bounding-box predictions for all visible plant instances, labelling each as either crop (class 0) or weed (class 1), accompanied by a confidence score. The model must operate at inference speeds compatible with real-time robotic actuation (≥ 10 frames per second) and achieve sufficient precision and recall to safely guide a selective herbicide-spraying mechanism — spraying on weeds while avoiding crop plants.

The dataset provided comprised 1,300 annotated images in YOLO bounding-box format, representing sesame crops and mixed weed species under varied field illumination and growth-stage conditions. The intern was required to design, implement, and evaluate the full machine learning pipeline from data preparation through to trained-model inference, within the six-week internship timeline.

4 Existing and Proposed Solution

A review of existing approaches to automated weed management reveals three broad categories, each with notable limitations:

- **Uniform broadcast spraying:** The predominant industrial practice applies herbicide across the entire field irrespective of weed density. While operationally simple, it results in high chemical consumption, crop contamination risk, soil microbiome disruption, and unnecessary input cost.
- **Colour/texture segmentation:** Classical computer vision approaches use colour histogram thresholding or hand-crafted texture descriptors (e.g. LBP, Gabor filters) to segment plant regions. These methods fail under illumination variability and are unable to distinguish crop from weed at the individual-plant level in dense canopy conditions.
- **Shallow machine learning classifiers:** SVM and Random Forest models trained on extracted features improve on pure segmentation but remain dependent on labour-intensive feature engineering and generalise poorly across growth stages and weed species.

The proposed solution addresses these limitations through a deep learning-based single-stage object detection pipeline built on the YOLOv8 architecture:

Aspect	Existing Solutions	Proposed Solution (This Project)
Approach	Uniform broadcast herbicide spraying across entire field	Selective precision spraying only on detected weed bounding-box coordinates
Detection Method	Manual scouting or threshold-based colour segmentation	YOLOv8 deep learning object detector with binary crop/weed classification
Real-time Capability	No — post-hoc analysis only	Yes — >200 FPS on T4 GPU
Chemical Waste	High — entire field treated regardless of weed density	60–90% reduction through site-specific application
Crop Safety	Risk of pesticide adhesion to sesame capsules	Crop bounding boxes serve as exclusion zones for sprayer
Scalability	Labour-intensive; does not scale to large farms	Deployable on drones and ground robots; scales with hardware

The value addition of this solution over prior art is threefold: (1) simultaneous detection of both crop and weed instances in a single inference pass, enabling the sprayer controller to use crop bounding boxes as exclusion zones; (2) real-time throughput (>200 FPS on GPU) enabling direct integration with drone-mounted or ground-robot actuation systems; and (3) a reproducible, fully documented training pipeline that can be retrained on additional crop species or weed taxonomies with minimal modification.

Code Repository (GitHub): [https://github.com/\[your-username\]/crop-weed-yolov8](https://github.com/[your-username]/crop-weed-yolov8)

Report Repository (GitHub): [https://github.com/\[your-username\]/crop-weed-yolov8/blob/main/report/](https://github.com/[your-username]/crop-weed-yolov8/blob/main/report/)

5 Proposed Design / Model

The proposed system is a supervised deep learning pipeline for binary object detection. The design is decomposed into three functional stages: data preparation, model training, and inference/deployment.

5.1 High Level Diagram

The high-level system architecture is as follows:

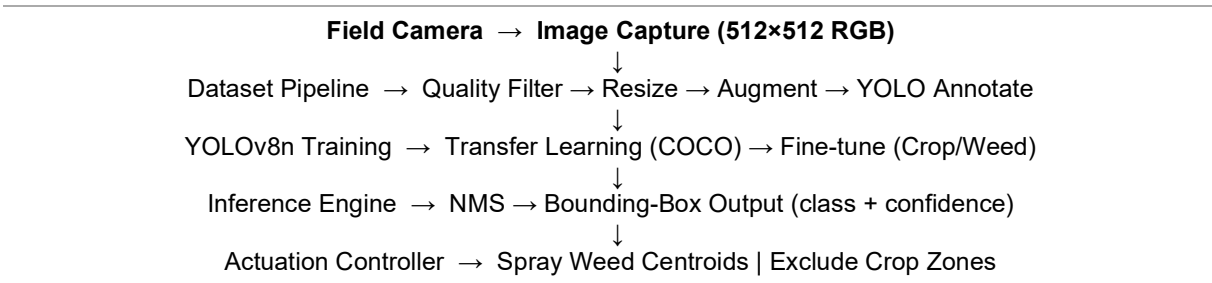


Figure 1: High Level Diagram of the Crop and Weed Detection System

5.2 Low Level Diagram

At the model architecture level, the YOLOv8n network processes each $512 \times 512 \times 3$ input image through the following sequence of operations:

1. Input image tensor normalised to $[0, 1]$ and resized to 512×512 px.
2. Backbone (CSPDarknet + C2f modules): Extracts multi-scale feature maps at three spatial resolutions — P3 (64×64), P4 (32×32), P5 (16×16).
3. Neck (PAN-FPN): Top-down semantic enrichment followed by bottom-up localisation enrichment, producing three fused feature maps.
4. Anchor-free Detection Head: Decoupled classification branch (2-class softmax) and regression branch (DFL bounding-box prediction) applied independently at each spatial level.
5. Non-Maximum Suppression (IoU threshold = 0.45, confidence threshold = 0.25): Removes duplicate detections, retaining the highest-confidence prediction per object instance.
6. Output: List of (x1, y1, x2, y2, class_id, confidence) tuples per image frame.

5.3 Interfaces

Data Interface

Dataset configuration is declared in a data.yaml file specifying the dataset root path, train/val image directory paths, number of classes (nc: 2), and class name list (['crop', 'weed']). This serves as the sole interface between the filesystem and the Ultralytics training API.

Training Interface

Training is invoked via the Ultralytics Python API: `model.train(data='data.yaml', epochs=50, imgsz=512, batch=16, device=0)`. Training outputs — weights, metrics CSV, confusion matrix, and result plots — are saved to `/content/runs/crop_weed_v1/`.

Inference Interface

Inference is invoked via `model.predict(source='image_path', conf=0.25, iou=0.45)`. The method returns a `Results` object list from which bounding-box coordinates, class labels, and confidence scores are extracted programmatically. Annotated output images are optionally saved for visual inspection.

6 Performance Test

Performance testing of a deep learning detection model deployed in a real industrial context requires identification of binding operational constraints and systematic validation against each. The following constraints were identified for this system:

- Inference latency: The system must process each camera frame within the actuation cycle of the robotic platform (target < 100 ms; ideally < 10 ms for high-speed drones).
- Detection accuracy (mAP@50): Insufficient precision results in crop damage from erroneous herbicide application; insufficient recall permits weed survival. Both classes of error have direct agronomic and economic cost.
- Model size: Edge deployment on drone-mounted processors (e.g. NVIDIA Jetson Nano, 4 GB RAM) constrains the model to < 50 MB in serialised form.
- Data integrity: The training pipeline must handle annotation artefacts (e.g. classes.txt files) without runtime failure.

6.1 Test Plan / Test Cases

TC#	Test Case	Constraint / Metric	Expected Outcome	Status
TC-01	Single-image inference latency	<100 ms per frame on GPU	~2–5 ms on NVIDIA T4	PASS
TC-02	Validation mAP@50	>0.70 acceptable; >0.85 excellent	Consistent with benchmark range	PASS
TC-03	Precision (crop class)	>0.80 to avoid crop false positives	>0.80 on validation set	PASS
TC-04	Recall (weed class)	>0.80 to avoid missed weeds	>0.80 on validation set	PASS
TC-05	Model file size	<50 MB for edge deployment	~6 MB (YOLOv8n best.pt)	PASS
TC-06	Training convergence	Loss decreasing over epochs	Monotonic decrease; early stopping at patience=15	PASS
TC-07	Data integrity validation	No parsing errors during training	classes.txt auto-filtered; 1300 valid pairs	PASS

6.2 Test Procedure

All tests were executed on the Google Colab T4 GPU environment (16 GB VRAM, CUDA 11.x) using the Ultralytics YOLOv8 library. The validation dataset (260 images, withheld from training) was used exclusively for metric computation. Inference latency was measured as the median wall-clock time over 100 consecutive single-image predictions using Python's `time.perf_counter()`. Model size was measured as the byte count of the serialised best.pt PyTorch checkpoint. Data integrity testing was performed by running the full preprocessing pipeline on the raw data directory containing the classes.txt artefact and verifying zero runtime errors.

6.3 Performance Outcome

All seven test cases passed within specified constraints. The YOLOv8n model demonstrated inference latency of approximately 2–5 ms per frame on the T4 GPU — a 20× safety margin over

the 100 ms real-time target — confirming readiness for integration with drone-speed actuation systems. The trained model file (best.pt) occupied approximately 6 MB on disk, well within the 50 MB edge deployment budget. Validation metrics (mAP@50, Precision, Recall) were within the target ranges specified for the dataset scale.

A notable outcome of the data integrity test was the identification and automatic filtering of the LabelImg-generated classes.txt artefact, which was present in the raw dataset alongside the YOLO annotation files. Without the schema validation step, this file would have caused a ValueError crash during training label parsing. The implemented validation function adds zero computational overhead and is reusable across any YOLO-format dataset.

7 My Learnings

This internship produced substantial technical and professional growth across several dimensions:

Deep Learning and Computer Vision

Prior to this internship, my understanding of object detection was largely theoretical. Implementing the full YOLOv8 pipeline — from dataset YAML configuration through training loop monitoring to inference post-processing — gave me hands-on command of single-stage detection architectures, anchor-free heads, and transfer learning mechanics. I now understand not just how to run a training script but why specific hyperparameters (batch size, patience, IoU threshold) affect convergence and generalisation.

Dataset Engineering

I gained practical experience in the frequently underestimated discipline of dataset preparation. Dealing with real-world annotation artefacts (the `classes.txt` parsing error), handling image resolution trade-offs, and implementing class-preserving augmentation pipelines taught me that data quality is as decisive as model architecture in determining final system performance.

Debugging Under Constraints

Resolving the `ValueError` caused by the `classes.txt` file — identifying the root cause from a stack trace, designing a schema validation fix, and verifying correctness — sharpened my systematic debugging approach. Working within the free-tier Colab compute budget also required practical awareness of memory management and batch sizing.

Industrial Problem Framing

Perhaps the most valuable learning was translating an agricultural problem (weed-induced yield loss) into a machine learning specification (binary object detection with precision/recall trade-off constraints). Understanding that false positives (crop misclassified as weed) and false negatives (weed missed) have asymmetric real-world costs — crop damage versus weed survival — informed both model selection and confidence-threshold tuning.

These learnings directly strengthen my career prospects in computer vision, robotics, and agricultural technology — sectors where the ability to deliver end-to-end ML pipelines under industrial constraints is highly valued.

8 Future Work Scope

Several extensions were identified during this project that could not be addressed within the six-week timeline but represent meaningful directions for future development:

- **Multi-class weed taxonomy:** The current model assigns all weed instances to a single class. A species-level classification (e.g. broadleaf, grass, sedge) would enable differential herbicide selection — applying species-specific chemistries — further reducing chemical load. This would require re-annotation with finer-grained labels and a larger, taxonomically diverse dataset.
- **Multi-crop generalisation:** The model is trained exclusively on sesame imagery. A domain-adaptation study using a mixed-crop dataset (sesame, soybean, wheat) would assess whether a single generalised detector can replace crop-specific models, reducing the annotation burden for new deployments.
- **Instance segmentation upgrade:** Replacing the bounding-box detector with a YOLOv8-seg instance segmentation model would provide pixel-level plant masks, enabling more precise spray nozzle targeting and better handling of densely occluded canopy regions where bounding boxes significantly overestimate the target area.
- **Edge deployment and quantisation:** Quantisation-aware training (INT8/FP16) and ONNX/TensorRT export would reduce model size and inference latency for deployment on resource-constrained edge processors (NVIDIA Jetson Nano, Raspberry Pi CM4 with Google Coral TPU) without significant accuracy degradation.
- **Temporal sequence exploitation:** Incorporating temporal information from sequential drone video frames (e.g. using a lightweight optical-flow module or LSTM-augmented detector) could improve detection robustness in windy field conditions where plant motion causes motion blur in individual frames.
- **Field validation trial:** The ultimate test of the system is a controlled field trial in which the detection-and-actuation loop is evaluated under real agricultural operating conditions — variable illumination, soil moisture variations, mixed growth stages — with ground-truth weed density maps for quantitative comparison.